



Program 2: Benchmark Case 1

Description

Benchmark 1: This program sequentially processes a dataset of 20 students. Each student record consists of 4 fields (ID, M1, M2, M3, M4). The program accumulates their totals iteratively, checks the total against a maximum mark cap of 100, and ultimately computes the overall average mark across all students.

Note: Each IN input will be a 32-bit value stored in a 32-bit register. It will go in order of all 20 IDs, all 20 M1s, and so on, in groups of 4.

Register Map:

- R1 = write pointer
- R2 = field counter
- R3 = ID
- R4 = M1
- R5 = M2
- R6 = M3
- R7 = M4
- R8 = Total
- R10 = Read IN
- R11 = Total DMEM offset
- R13 = Processed records
- R14 = DMEM offset pointer
- R15 = Current total
- R16 = Sum of totals
- R17 = Num records
- R18 = Average of totals
- R28 = total students (20)
- R29 = total count (100)
- R30 = Mark Cap (100)
- R1 = write pointer
- R2 = field counter
- R3 = ID
- R4 = M1
- R5 = M2
- R6 = M3
- R7 = M4
- R8 = Total
- R30 = Mark Cap (100)
- R29 = total count (100)
- R28 = total students (20)

- R10 = Read IN
- R11 = Total DMEM offset
- R13 = Processed records
- R14 = DMEM offset pointer
- R15 = Current total
- R16 = Sum of totals
- R17 = Num records
- R18 = Average of totals

Assembly Code	Binary (Hex)
ADDI R28, R28, 20	279C0014
ADDI R29, R29, 100 (100 = 20 students * 5 values each)	27BD0064
ADDI R30, R30, 100	27DE0064
Read_Loop:	
IN R10	300A0000
STORE R10, 0(R1)	202A0000
ADDI R1, R1, 1	24210001
ADDI R2, R2, 1	24420001
NOOP	00000000
NOOP	00000000
NOOP	00000000
BLT R2, R29, Read_Loop	2C5DFFF8
ADD R11, R1, R0	00205800
ADD R1, R0, R0	00000800
ADD R2, R0, R0	00001000
Proc_Loop:	
LOAD R3, 0(R1)	1C230000
LOAD R4, 1(R1)	1C240001
LOAD R5, 2(R1)	1C250002
LOAD R6, 3(R1)	1C260003
LOAD R7, 4(R1)	1C270004
ADD R8, R4, R5	00854000
ADD R8, R8, R6	01064000
ADD R8, R8, R7	01074000
NOOP	00000000
NOOP	00000000
NOOP	00000000
BLT R8, R30, Cap_Ok	2D1E0001
ADD R8, R30, R0	03C04000
Cap_Ok:	
ADD R14, R11, R13	016D7000
STORE R8, 0(R14)	21C80000
ADDI R13, R13, 1	25AD0001
Next_ID:	

Assembly Code		Binary (Hex)
ADD	R1, R1, 5	00250800
NOOP		00000000
NOOP		00000000
NOOP		00000000
BLT	R13, R28, Proc_Loop	2DBCFFEB
ADD	R1, R0, R0	00000800
Avg_Totals:		
ADD	R14, R11, R1	01617000
LOAD	R15, 0(R14)	1DCF0000
ADD	R16, R16, R15	020F8000
ADDI	R17, R17, 1	26310001
NOOP		00000000
NOOP		00000000
NOOP		00000000
BLT	R17, R28, Avg_Totals	2E3CFFF8
DIV	R18, R16, R28	1A1C9000
OUT	R18	36400000

